

Felipe Nunes

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June 1, 2025

Hiring Committee – Requisition #16668
Ansys Canada
Vancouver, British Columbia

Re: Spring 2026 Electronics Simulation R&D Intern

Dear Hiring Committee,

I am an Integrated Engineering student who lives for the moment when a line of code turns raw hardware into something intelligent and alive. Ansys' mission, bridging ideas and reality through world-class simulation, mirrors the challenge I chase every day: writing fast, reliable software that lets physical systems speak in numbers. The opportunity to join your Electronics Simulation R&D team feels like the perfect next step.

At Miru Smart Technologies, I built a custom CMake-linker-script toolchain for an ARM64 target that trimmed compile times and enforced rigorous static analysis, all while shrinking a mixed-signal PCB by 60 % and driving electrochromic windows with new Zephyr C modules that boosted transition speed by 50 %. That experience taught me that elegant firmware and disciplined build systems are inseparable from great hardware.

Previously, at Sarcomere Dynamics, I crafted a modular C++ actuator-control library and extended the company's Python API to stream telemetry over Wi-Fi, enabling real-time tuning of a multi-DOF robotic hand. Integrating finite-state machines, USB-C power delivery, and mixed-signal diagnostics pushed me to design unit tests that caught edge-case race conditions before the first prototype left the bench.

My capstone, a robotic chessboard, deepened this software focus. I compiled a custom GRBL fork, built hardware-abstraction layers, and wrote a polling algorithm that tracks 64 Hall-effect sensors with microsecond-level determinism, earning our faculty's Best Engineering Project award. Whether I am shaping FreeRTOS tasks for a CAN-bus HUD or scripting regression tests for sensor-fusion code, I am happiest translating physics into performant, maintainable software.

I want to bring that energy to Ansys. Your team's work on high-performance electromagnetics and circuit solvers is exactly where my passions converge: numerical methods, rigorous testing, and software craftsmanship with real-world impact. I thrive in collaborative code reviews, and your ONE Ansys values of courage, generosity, and outstanding results resonate with how I strive to lead student teams and mentor peers.

I would love to discuss how my blend of embedded-systems insight and software-driven problem-solving can contribute to the next generation of Ansys simulation tools. Thank you for your time and consideration and I look forward to the possibility of working together.

Sincerely,
Felipe Nunes